

Familial partial lipodystrophy

Description

Familial partial lipodystrophy is a rare condition that is characterized by the loss of fatty (adipose) tissue that is found underneath the skin (subcutaneous). Affected individuals typically lose subcutaneous adipose tissue from the arms; legs; and the truncal region, which includes the chest, abdomen, and hips. Familial partial lipodystrophy is part of a group of disorders known as lipodystrophies, which can cause the loss of varying amounts of adipose tissue. The severity of familial partial lipodystrophy and the specific signs and symptoms that are seen in affected individuals can vary, even among members of the same family.

Adipose tissue is found in many parts of the body. It stores fat for energy and provides support for the body's structures. In people with familial partial lipodystrophy, the loss of subcutaneous adipose tissue from the arms, legs, and hips gives these parts of the body a muscular appearance. The fat that is normally stored in these areas builds up in other parts of the body, including the face, neck, and inside the abdomen. This abnormal fat distribution often becomes apparent during childhood or around the time of puberty.

Abnormal fat storage in the body can lead to health problems throughout a person's life. Many people with familial partial lipodystrophy develop insulin resistance, a condition in which the body's tissues cannot adequately respond to insulin. Insulin is a hormone that helps regulate levels of blood glucose, also called blood sugar. Insulin resistance may lead to a condition called diabetes mellitus. People with this condition have high levels of blood glucose.

Additional signs and symptoms of familial partial lipodystrophy can include:

- A skin condition related to high levels of insulin in which the skin in the folds and creases of the body becomes thick, dark, and velvety (acanthosis nigricans)
- High levels of fats called triglycerides in the bloodstream (hypertriglyceridemia), which can cause inflammation of the pancreas (acute pancreatitis)
- An enlarged liver (hepatomegaly) due to an abnormal buildup of fats (hepatic steatosis)

Some affected individuals develop high blood pressure and a buildup of fatty deposits in

the arteries (atherosclerosis), which can increase the risk of a form of heart disease called coronary artery disease. After puberty, some women with familial partial lipodystrophy develop signs and symptoms that are likely related to hormonal changes. These can include:

- Multiple cysts on the ovaries
- An increased amount of body hair (hirsutism)
- An inability to conceive (infertility)

There are several forms of familial partial lipodystrophy, which are distinguished by their genetic cause. The most common form is type 2, also called Dunnigan disease. In addition to the signs and symptoms described above, people with familial partial lipodystrophy type 2 may develop:

- Weakness in the muscles used for movement (skeletal muscles)
- Abnormalities of the heart muscle (cardiomyopathy)
- Irregular heartbeats (arrhythmias)

In people with familial partial lipodystrophy type 2, excess fat may build up around the face, neck, and upper back, especially in women. This appearance is sometimes described as "cushingoid," because it resembles the appearance of people with a hormonal disorder called Cushing syndrome.

Frequency

Familial partial lipodystrophy is a rare disease that affects 1 in 7,000 to 2.3 in 1 million people. However, the exact prevalence is difficult to determine, as many people with this condition likely do not receive a diagnosis. Women tend to receive a diagnosis of familial partial lipodystrophy more often than men. This may be because the loss of fat from the arms, legs, and hips is more easily recognized in women or because complications, such as diabetes mellitus and hypertriglyceridemia, tend to be more severe in women.

Causes

Genetic changes that cause disease are called pathogenic variants. Pathogenic variants in several different genes cause familial partial lipodystrophy. Pathogenic variants in the lamin A/C (*LMNA*) gene cause familial partial lipodystrophy type 2. The other forms of the disorder are caused by pathogenic variants in other genes.

The *LMNA* gene provides instructions for making proteins called lamins, which are an important part of the structure that surrounds the nucleus inside cells (the nuclear envelope). Through a mechanism that is not well understood, pathogenic variants in the *LMNA* gene appear to lead to the premature death of adipocytes, which are the fat-

storing cells in adipose tissue. The loss of these cells disrupts the body's ability to store and use fats properly. This leads to the abnormal distribution of adipose tissue and the resulting health complications that are characteristic of this disorder. Changes in adipose tissue alter hormone production and affect many of the body's organs. However, it is unclear exactly why these changes cause fat to be lost in some parts of the body and abnormally stored in others.

The other genes that are associated with familial partial lipodystrophy provide instructions for making proteins with important roles in fat storage. Specifically, these proteins are necessary for the proper development and function of adipocytes. Pathogenic variants in any of the genes that are associated with familial partial lipodystrophy impair the development, structure, or function of adipocytes and make the body unable to store and use fats properly.

In some people with familial partial lipodystrophy, a pathogenic variant cannot be identified. In these cases, the cause of the disorder is unknown.

[Learn more about the genes associated with Familial partial lipodystrophy](#)

- CAV1
- LMNA
- NOTCH3

Additional Information from NCBI Gene:

- ACAA2
- ADRA2A
- AKT2
- CIDEA
- LIPE
- PCYT1A
- PLAAT3
- PLIN1
- PPARG

Inheritance

Most cases of familial partial lipodystrophy, including type 2, are inherited in an autosomal dominant pattern, which means one copy of the altered gene in each cell is sufficient to cause the disorder. In some cases, an affected person inherits the pathogenic variant from a parent. Other cases result from a new (de novo) pathogenic variant in the gene that occurs during the formation of reproductive cells (eggs or sperm) in an affected individual's parent or during early embryonic development. These affected individuals typically have no history of the disorder in their family.

Some types of familial partial lipodystrophy are inherited in an autosomal recessive pattern, which means both copies of the gene in each cell have a pathogenic variant. The parents of an individual with an autosomal recessive condition each carry one copy of the altered gene, but they typically do not show signs and symptoms of the condition.

Other Names for This Condition

- Dunnigan-Kobberling syndrome
- FPL
- Kobberling-Dunnigan syndrome
- Lipodystrophy, familial partial

Additional Information & Resources

Genetic Testing Information

- Genetic Testing Registry: Familial partial lipodystrophy (<https://www.ncbi.nlm.nih.gov/gtr/conditions/C0271694/>)
- Genetic Testing Registry: AKT2-related familial partial lipodystrophy (<https://www.ncbi.nlm.nih.gov/gtr/conditions/C5680134/>)
- Genetic Testing Registry: CIDEC-related familial partial lipodystrophy (<https://www.ncbi.nlm.nih.gov/gtr/conditions/C3808940/>)
- Genetic Testing Registry: Familial partial lipodystrophy, Dunnigan type (<https://www.ncbi.nlm.nih.gov/gtr/conditions/C1720860/>)
- Genetic Testing Registry: Familial partial lipodystrophy, Kobberling type (<https://www.ncbi.nlm.nih.gov/gtr/conditions/C1720859/>)
- Genetic Testing Registry: PLIN1-related familial partial lipodystrophy (<https://www.ncbi.nlm.nih.gov/gtr/conditions/C5191005/>)
- Genetic Testing Registry: PPARG-related familial partial lipodystrophy (<https://www.ncbi.nlm.nih.gov/gtr/conditions/C1720861/>)

Genetic and Rare Diseases Information Center

- Familial partial lipodystrophy (<https://rarediseases.info.nih.gov/diseases/11962/index>)
- Familial partial lipodystrophy, Dunnigan type (<https://rarediseases.info.nih.gov/diseases/3126/index>)
- Familial partial lipodystrophy, Köbberling type (<https://rarediseases.info.nih.gov/diseases/12598/index>)
- PPARG-related familial partial lipodystrophy (<https://rarediseases.info.nih.gov/diseases/12600/index>)
- CIDEC-related familial partial lipodystrophy (<https://rarediseases.info.nih.gov/diseases/12600/index>)

es/13125/index)

- AKT2-related familial partial lipodystrophy (<https://rarediseases.info.nih.gov/diseases/12599/index>)
- PLIN1-related familial partial lipodystrophy (<https://rarediseases.info.nih.gov/diseases/12601/index>)
- LIPE-related familial partial lipodystrophy (<https://rarediseases.info.nih.gov/diseases/13126/index>)

Patient Support and Advocacy Resources

- National Organization for Rare Disorders (NORD) (<https://rarediseases.org/>)

Clinical Trials

- ClinicalTrials.gov ([https://clinicaltrials.gov/search?cond=%22Familial partial lipodystrophy%22](https://clinicaltrials.gov/search?cond=%22Familial%20partial%20lipodystrophy%22))

Catalog of Genes and Diseases from OMIM

- LIPODYSTROPHY, FAMILIAL PARTIAL, TYPE 1; FPLD1 (<https://omim.org/entry/608600>)
- LIPODYSTROPHY, FAMILIAL PARTIAL, TYPE 2; FPLD2 (<https://omim.org/entry/151660>)
- LIPODYSTROPHY, FAMILIAL PARTIAL, TYPE 3; FPLD3 (<https://omim.org/entry/604367>)
- LIPODYSTROPHY, FAMILIAL PARTIAL, TYPE 4; FPLD4 (<https://omim.org/entry/613877>)
- LIPODYSTROPHY, FAMILIAL PARTIAL, TYPE 5; FPLD5 (<https://omim.org/entry/615238>)
- LIPODYSTROPHY, FAMILIAL PARTIAL, TYPE 6; FPLD6 (<https://omim.org/entry/615980>)
- LIPODYSTROPHY, FAMILIAL PARTIAL, TYPE 7; FPLD7 (<https://omim.org/entry/606721>)
- LIPODYSTROPHY, FAMILIAL PARTIAL, TYPE 8; FPLD8 (<https://omim.org/entry/620679>)
- LIPODYSTROPHY, FAMILIAL PARTIAL, TYPE 9; FPLD9 (<https://omim.org/entry/620683>)

Scientific Articles on PubMed

- PubMed (<https://pubmed.ncbi.nlm.nih.gov/?term=%28familial+partial+lipodystrophy>)

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